

Business Insights & Recommendations Document

Disaster Risk & Infrastructure Resilience Analysis (2015–2024)

1. Executive Summary

This project analyzes disaster events, infrastructure exposure, population vulnerability, economic losses, and recovery costs from 2015–2024 to identify locations and infrastructure assets requiring stronger disaster resilience.

The analysis focuses on eight key business questions:

- **Geographic concentration** of disaster risk
- Infrastructure damage by disaster type
- Relationship between disaster severity and economic loss
- Population vulnerability and human/economic impact
- Year-over-year disaster and loss trends
- Infrastructure age and damage vulnerability
- **Risk-based infrastructure** investment prioritization
- Recovery cost and slow-recovery risk

Objective: Transform historical disaster data into actionable resilience and investment insights, moving from hazard/exposure analysis toward prioritizing resilience investments.

2. Key Business Questions & Insights

Q1. Which states/districts show the highest combined score of hazard frequency, hazard severity, and critical infrastructure exposure?

Analysis: Composite Geographic Risk Measure

Geographic Risk Score = Hazard Frequency + Hazard Severity + Infrastructure Exposure

Locations are categorized into: **Very High Risk**, **High Risk**, **Moderate Risk**, and **Low Risk**.

Business Insight: Locations with simultaneously high disaster frequency, high severity, and substantial critical infrastructure exposure represent concentrated resilience risk affecting essential services and economic activity.

Recommendation: Prioritize high-risk districts for infrastructure reinforcement, early-warning systems, emergency response capacity, backup utilities, and disaster-resilient construction standards.

3. Infrastructure Damage by Disaster Type

Q2. Which infrastructure types sustain the greatest damage from each disaster type?

Disaster Type	Infrastructure Type	Primary Risk Impact
Flood	Bridges, Hospitals, Power Substations	Transportation, Water, Electrical
Cyclone	Power Infrastructure	Power & Communication
Earthquake	Buildings, Bridges, Utilities	Structural Integrity
Landslide	Roads & Transportation	Access Routes

Recommendation: Develop hazard-specific resilience plans rather than applying the same infrastructure strategy to every disaster.

4. Disaster Severity, Duration and Economic Loss

Q3. How strongly does disaster severity and duration correlate with infrastructure damage and total economic loss?

Correlation Coefficient	Interpretation
0.00 – 0.19	Very weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very strong

Recommendation: Use severity and duration as risk-model variables when planning emergency resources, infrastructure redundancy, insurance financing, and recovery budgets.

5. Population Vulnerability

Q4. Does population density or vulnerable population size correlate with fatalities and economic loss?

Business Insight: High exposure paired with a large vulnerable population faces substantially greater social consequences than sparsely populated zones.

Recommendation: Direct evacuation planning, shelters, public warnings, and medical capacity to high-population-risk zones.

6. Disaster Trend: 2015–2024

Q5. How has disaster frequency and economic loss trended year-over-year?

Calculation: $\text{YoY Growth} = (\text{Current Year} - \text{Previous Year}) / \text{Previous Year} \times 100$

Recommendation: Leverage 10-year trends to support long-term resilience budgets, insurance/risk financing, and mitigation programs.

7. Infrastructure Age Analysis

Q6. Is infrastructure age related to damage severity or criticality rating?

Infrastructure Age		Risk Group
0 – 10 years		New
11 – 20 years		Medium age
21 – 40 years		Older
40+ years		Very old

Recommendation: Conduct structural inspections and upgrade vulnerable components on older high-criticality assets.

8. Composite Risk Score & Investment Prioritization

Q7. Which locations should be prioritized first for infrastructure resilience investment?

Normalized Risk Score Formula:

- 30% Hazard Score
- 25% Infrastructure Exposure
- 20% Infrastructure Damage
- 15% Economic Loss
- 10% Population Vulnerability

Risk Score Range	Priority Level
80 – 100	Critical Priority

Risk Score Range	Priority Level
60 – 79	High Priority
40 – 59	Moderate Priority
0 – 39	Lower Priority

9. Recovery Cost Analysis

Q8. Which disaster types generate the highest recovery cost relative to initial damage?

Recovery Cost Ratio = Total Recovery Cost ÷ Initial Infrastructure Damage

Disasters with lower initial physical loss but high recovery ratios represent **slow-recovery risk** and ongoing financial strain.

10. Recommended Business Actions & Final Framework

1. **Protect Critical Infrastructure:** Prioritize high-criticality & high-hazard assets.
2. **Build Hazard-Specific Resilience:** Tailor responses for Floods, Cyclones, Earthquakes, etc.
3. **Upgrade Aging Infrastructure:** Target older critical assets.
4. **Strengthen Emergency Preparedness:** Focus on early warnings and evacuation planning.
5. **Establish Recovery-Financing:** Pre-position reconstruction funds and reserves.
6. **Monitor Risk Annually:** Track parameters continuously via a unified decision dashboard.